

Smith Chart



Transmission line calculation-such as the determination of input impedance, reflection coefficient and load impedance often involve tedious manipulations of complex numbers.

Graphical method of solution → **Smith Chart**

Smith Chart was devised by P.H.Smith (in 1930).

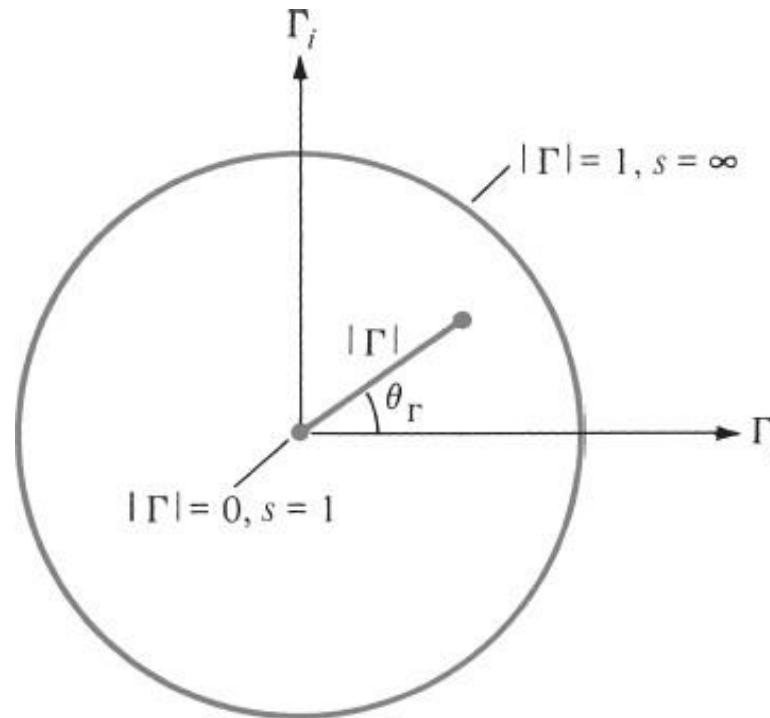
Smith Chart is a graphical plot of normalized resistance and reactance functions in the reflection-coefficient plane

Reflection Co-efficient plane (Finite plane)



$$\left[\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \right]$$

Z_0 normalized impedance



What
about
Z impedance
plane??
==

The **normalized impedance** is represented on the **Smith chart** by using families of curves that identify the **normalized resistance** r (real part) and the **normalized reactance** x (imaginary part)

Smith Chart Circle Equations



- If a lossless line of Z_0 is terminated with Z_L ,
 $z_L = Z_L/Z_0$ (normalized load impedance),

$$\Gamma = \frac{z_L - 1}{z_L + 1} = |\Gamma| e^{j\theta} \quad \longrightarrow \quad z_L = \frac{1 + |\Gamma| e^{j\theta}}{1 - |\Gamma| e^{j\theta}}$$

- Let $\Gamma = \Gamma_r + j\Gamma_i$, and $z_L = r_L + jx_L$.

$$r_L + jx_L = \frac{(1 + \Gamma_r) + j\Gamma_i}{(1 - \Gamma_r) - j\Gamma_i} \quad \longrightarrow \quad r_L = \frac{1 - \Gamma_r^2 - \Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2}$$

$$x_L = \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2}$$

$$\left(\Gamma_r - \frac{r_L}{1 + r_L} \right)^2 + \Gamma_i^2 = \left(\frac{1}{1 + r_L} \right)^2$$

r circle

$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L} \right)^2 = \left(\frac{1}{x_L} \right)^2$$

x circle

Smith Chart Circle Equations



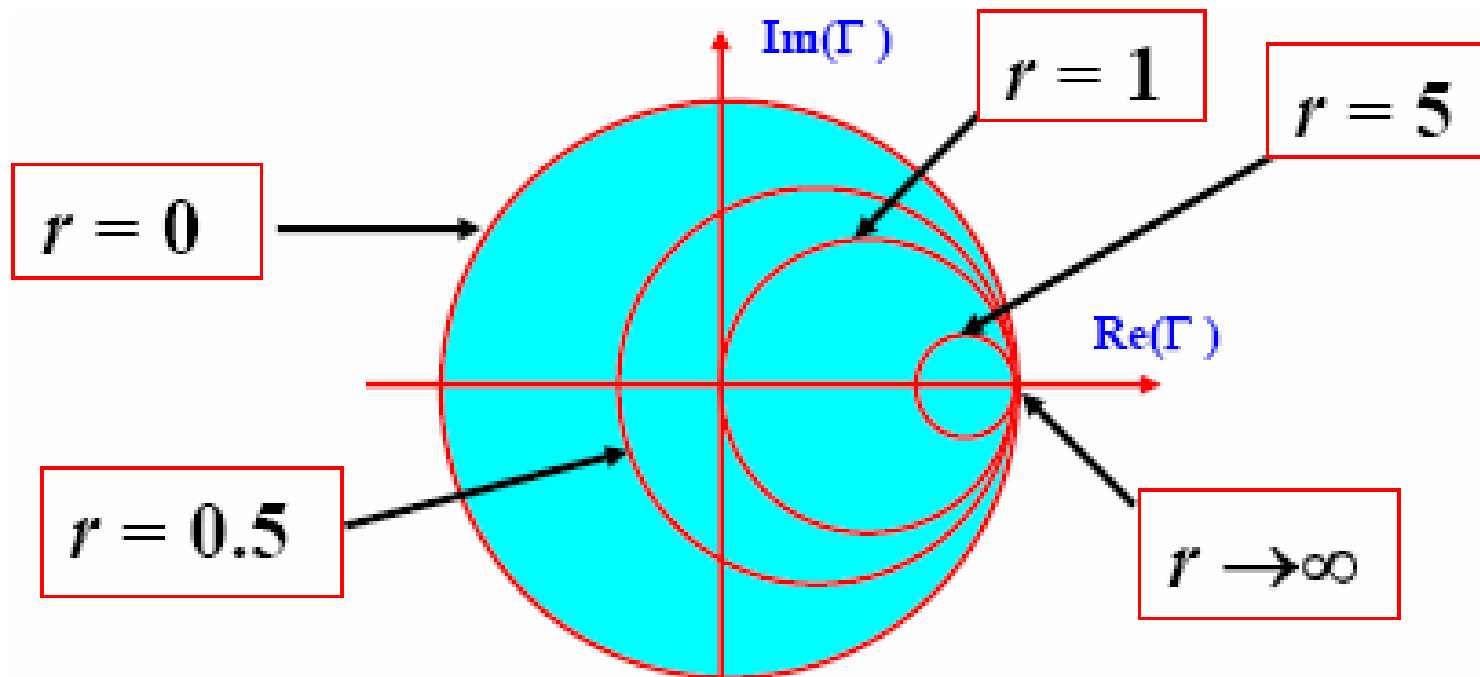
$$\left(\Gamma_r - \frac{r_L}{1 + r_L} \right)^2 + \Gamma_i^2 = \left(\frac{1}{1 + r_L} \right)^2$$
$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L} \right)^2 = \left(\frac{1}{x_L} \right)^2$$

The result for the **real part** indicates that on the complex plane with coordinates $(\text{Re}(\Gamma), \text{Im}(\Gamma))$ all the possible impedances with a given normalized resistance r are found on a **circle** with

$$\text{Center} = \left\{ \frac{r}{1 + r}, 0 \right\} \qquad \text{Radius} = \frac{1}{1 + r}$$

As the normalized resistance r varies from **0** to ∞ , we obtain a family of circles completely contained inside the domain of the reflection coefficient $|\Gamma| \leq 1$.

r-circles



$$\text{Center} = \left\{ \frac{r}{1+r}, 0 \right\}$$

$$\text{Radius} = \frac{1}{1+r}$$

Smith Chart Circle Equations



$$\left(\Gamma_r - \frac{r_L}{1 + r_L} \right)^2 + \Gamma_i^2 = \left(\frac{1}{1 + r_L} \right)^2,$$
$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L} \right)^2 = \left(\frac{1}{x_L} \right)^2$$

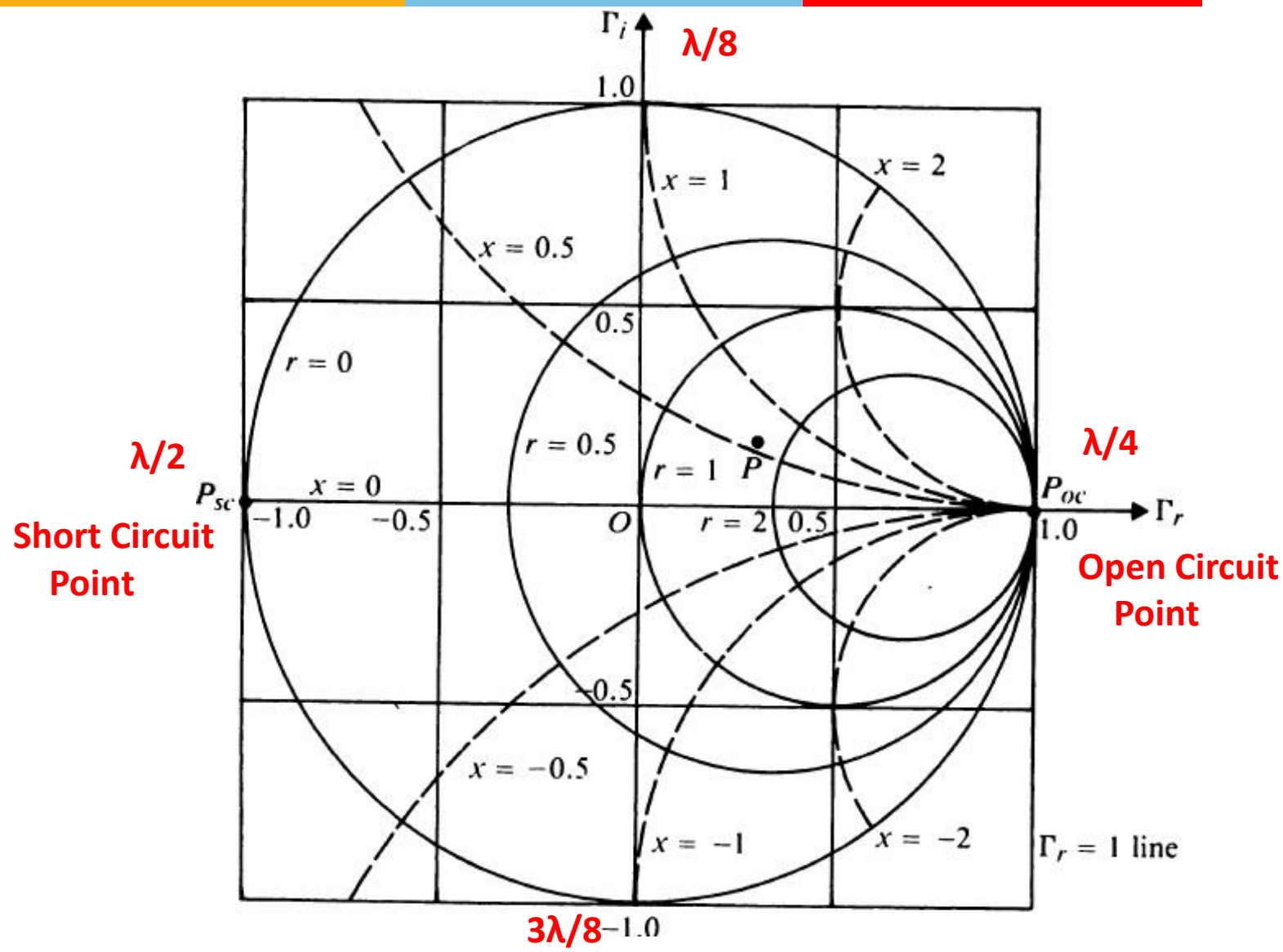
The result for the **imaginary part** indicates that on the complex plane with coordinates $(\text{Re}(\Gamma), \text{Im}(\Gamma))$ all the possible impedances with a given normalized reactance **x** are found on a **circle** with

$$\text{Center} = \left\{ 1, \frac{1}{x} \right\} \quad \text{Radius} = \frac{1}{x}$$

As the normalized reactance **x** varies from $-\infty$ to ∞ , we obtain a family of arcs contained inside the domain of the reflection coefficient **$|\Gamma| \leq 1$** .



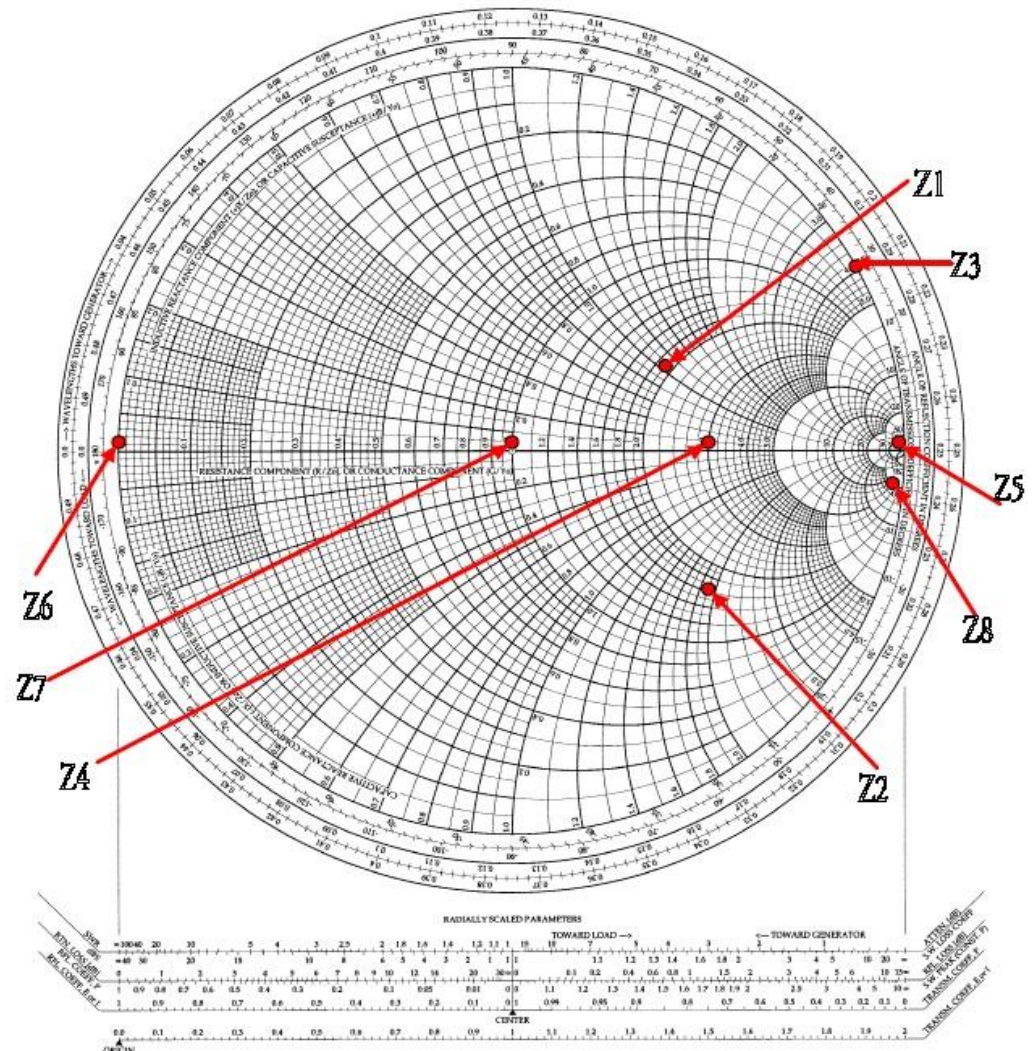
Smith Chart



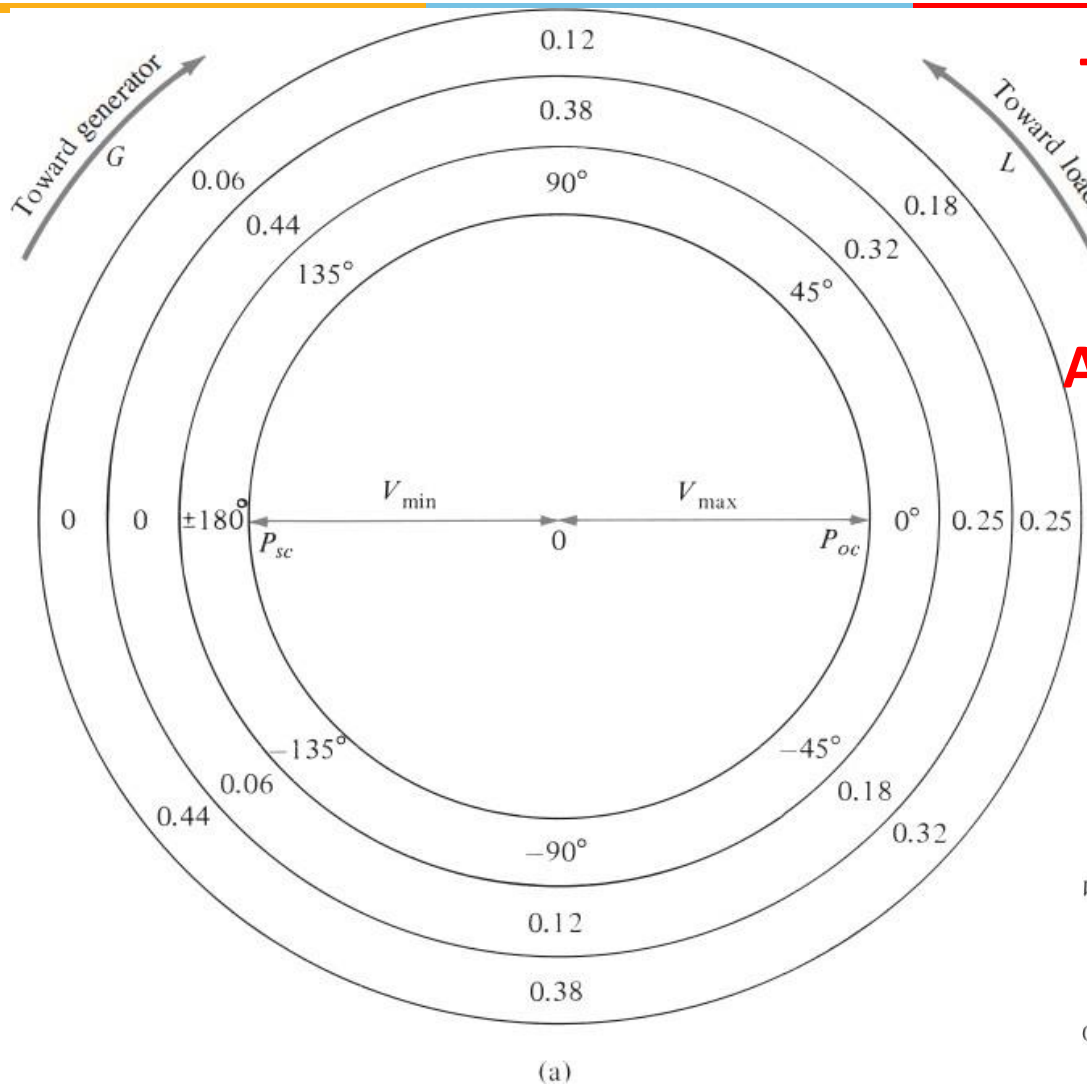
Important points about Smith chart



- Impedance divided by line impedance (50 Ohms)
 - $Z1 = 100 + j50$
 - $Z2 = 75 - j100$
 - $Z3 = j200$
 - $Z4 = 150$
 - $Z5 = \text{infinity}$ (an open circuit)
 - $Z6 = 0$ (a short circuit)
 - $Z7 = 50$
 - $Z8 = 184 - j900$
- Then, normalize and plot. The points are plotted as follows:
 - $z1 = 2 + j$
 - $z2 = 1.5 - j2$
 - $z3 = j4$
 - $z4 = 3$
 - $z5 = \text{infinity}$
 - $z6 = 0$
 - $z7 = 1$
 - $z8 = 3.68 - j18S$

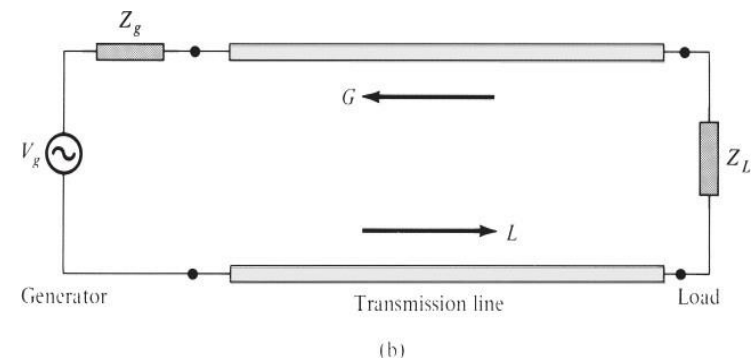


Important points about Smith chart



Towards Load: Anticlockwise

Away from Load: Clockwise

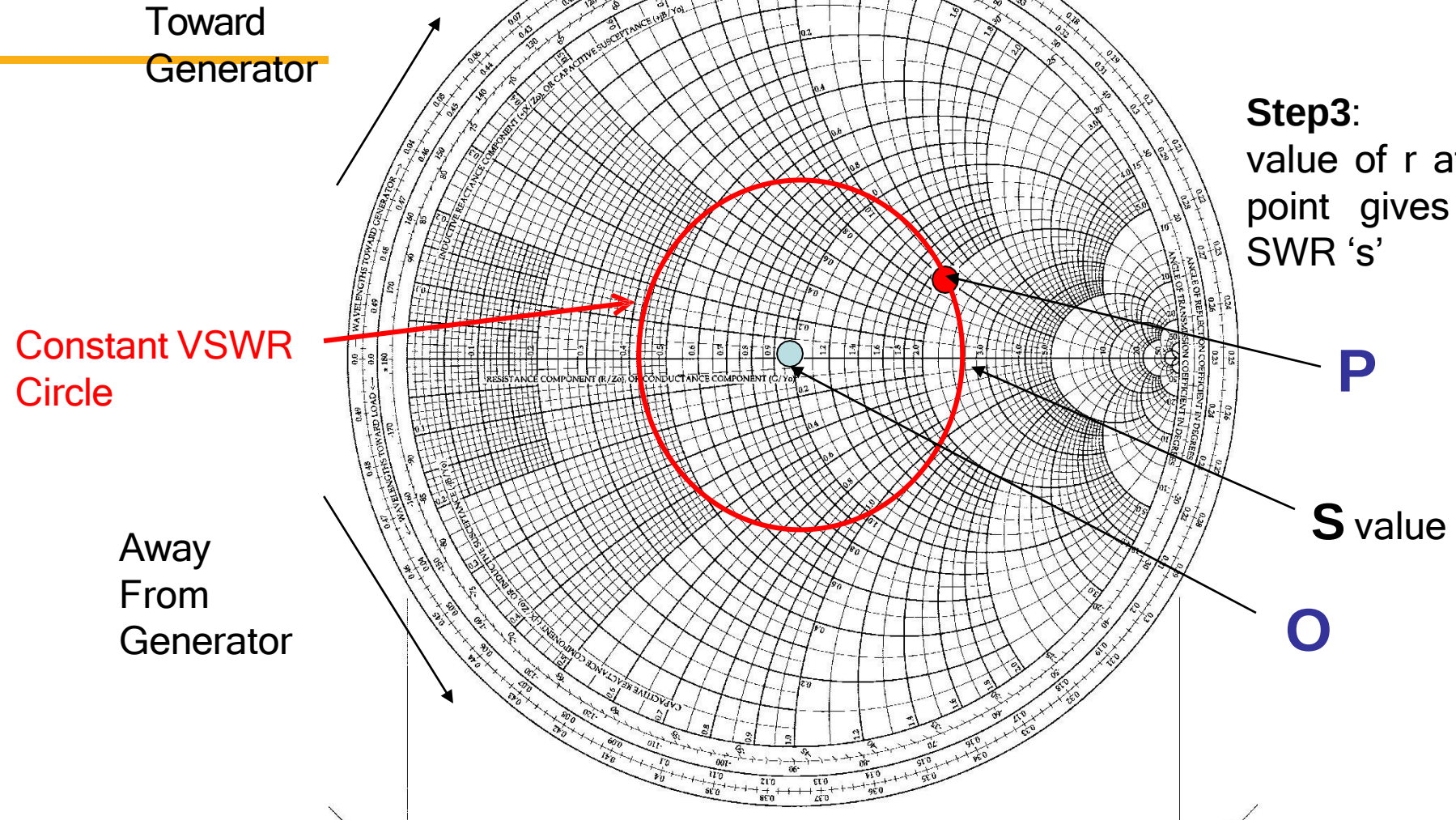


Plotting a constant VSWR circle

Step1: Locate the normalized load impedance on the chart (say P)

achieve lead

Step3: The value of r at this point gives you SWR 's'

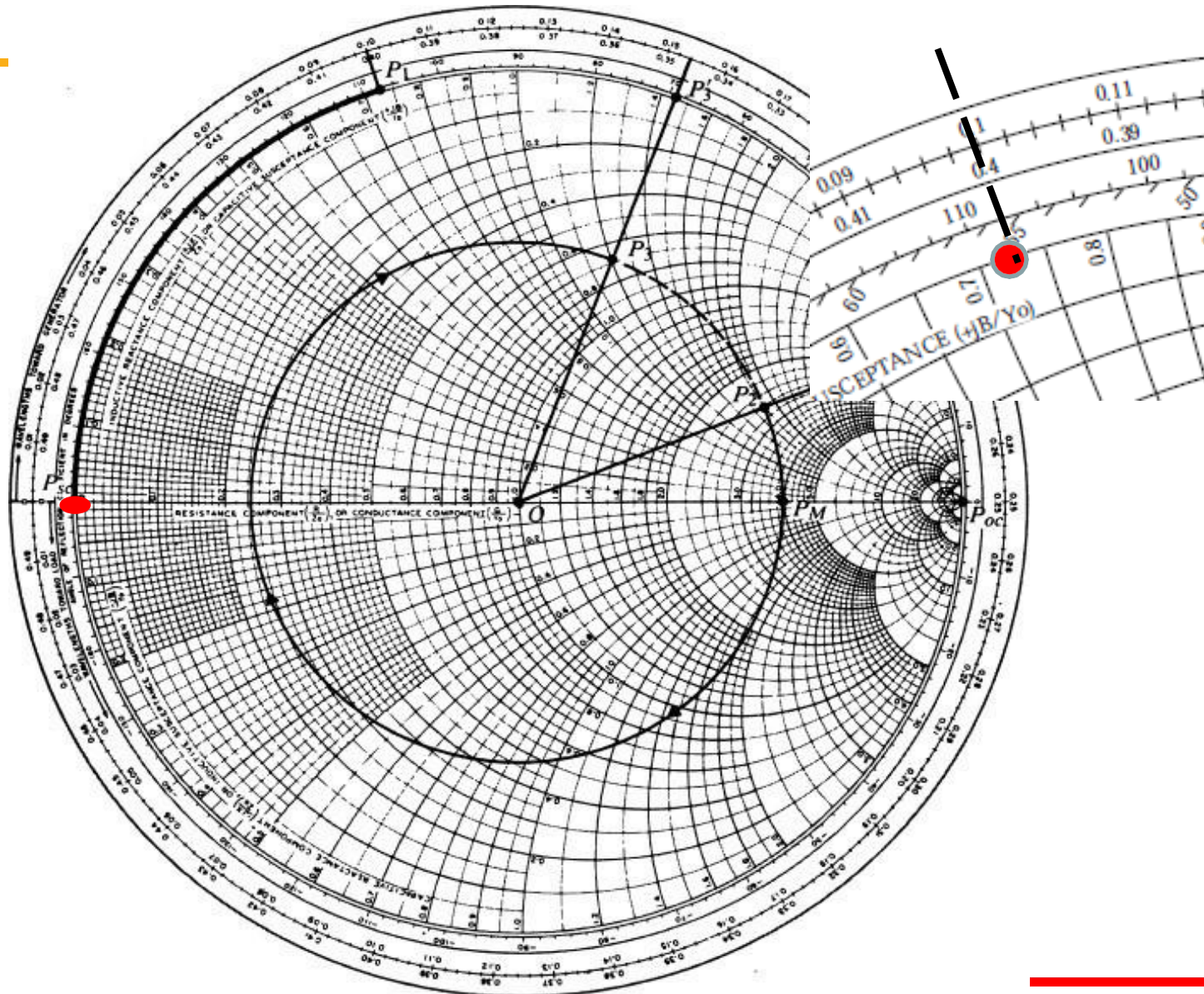


Step2: Take center of the Smith chart (O) as the center and the radius as OP and draw a circle. (The circle passes through the normalized load impedance).

Learning Example



Find the input impedance of a section of a $50\ \Omega$ loss-less transmission line that is 0.1λ long and is terminated in a short circuit.



To find the reflection coefficient for a known load

1. Normalization

$$\begin{aligned} z_n &= (25 + j\ 100)/50 \\ &= 0.5 + j\ 2.0 \end{aligned}$$

3. Find normalized reactance arc

$$x = 2.0$$

2. Find normalized resistance circle

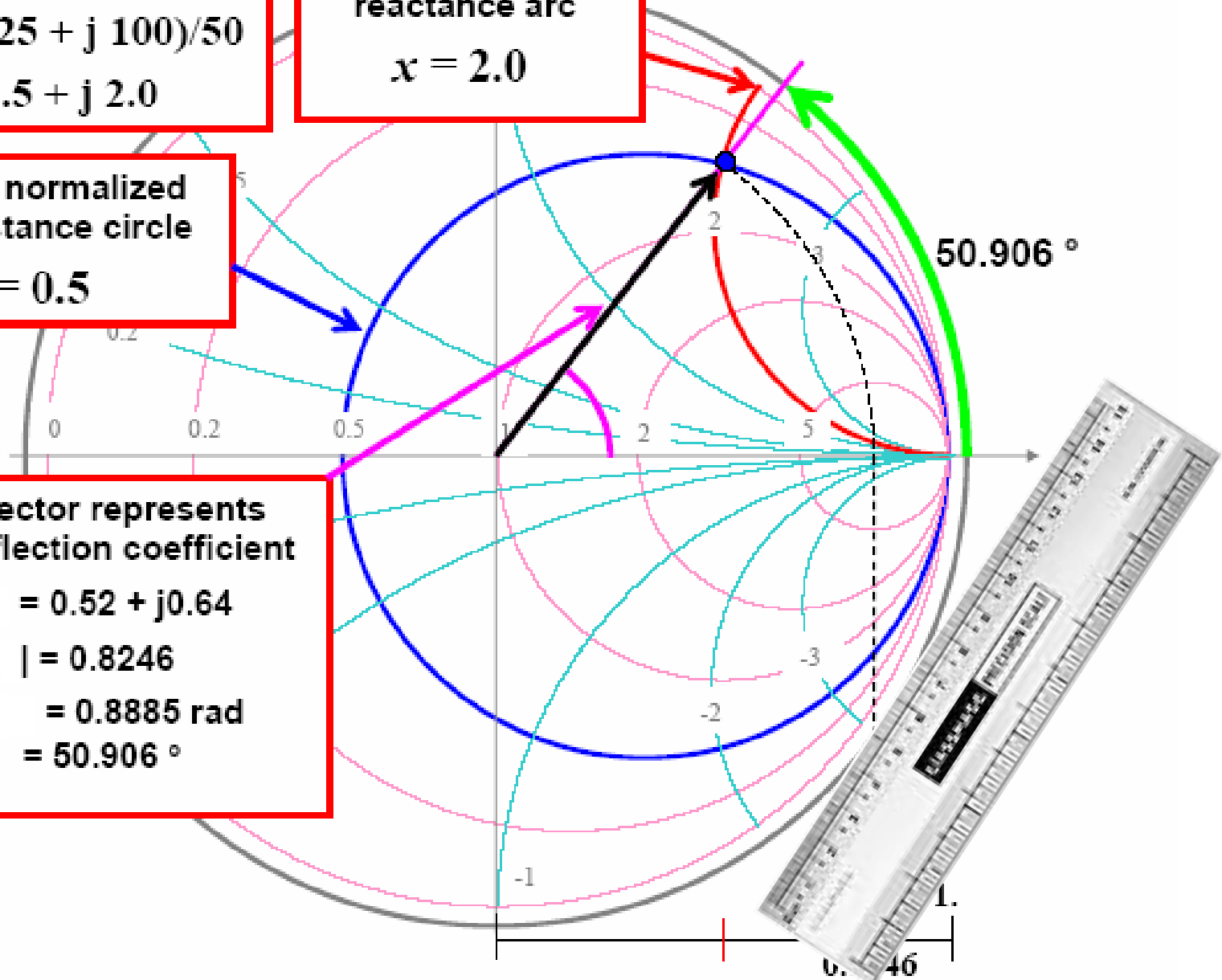
$$r = 0.5$$

4. This vector represents the reflection coefficient

$$\Gamma = 0.52 + j0.64$$

$$|\Gamma| = 0.8246$$

$$\begin{aligned} \angle \Gamma &= 0.8885 \text{ rad} \\ &= 50.906^\circ \end{aligned}$$



Example

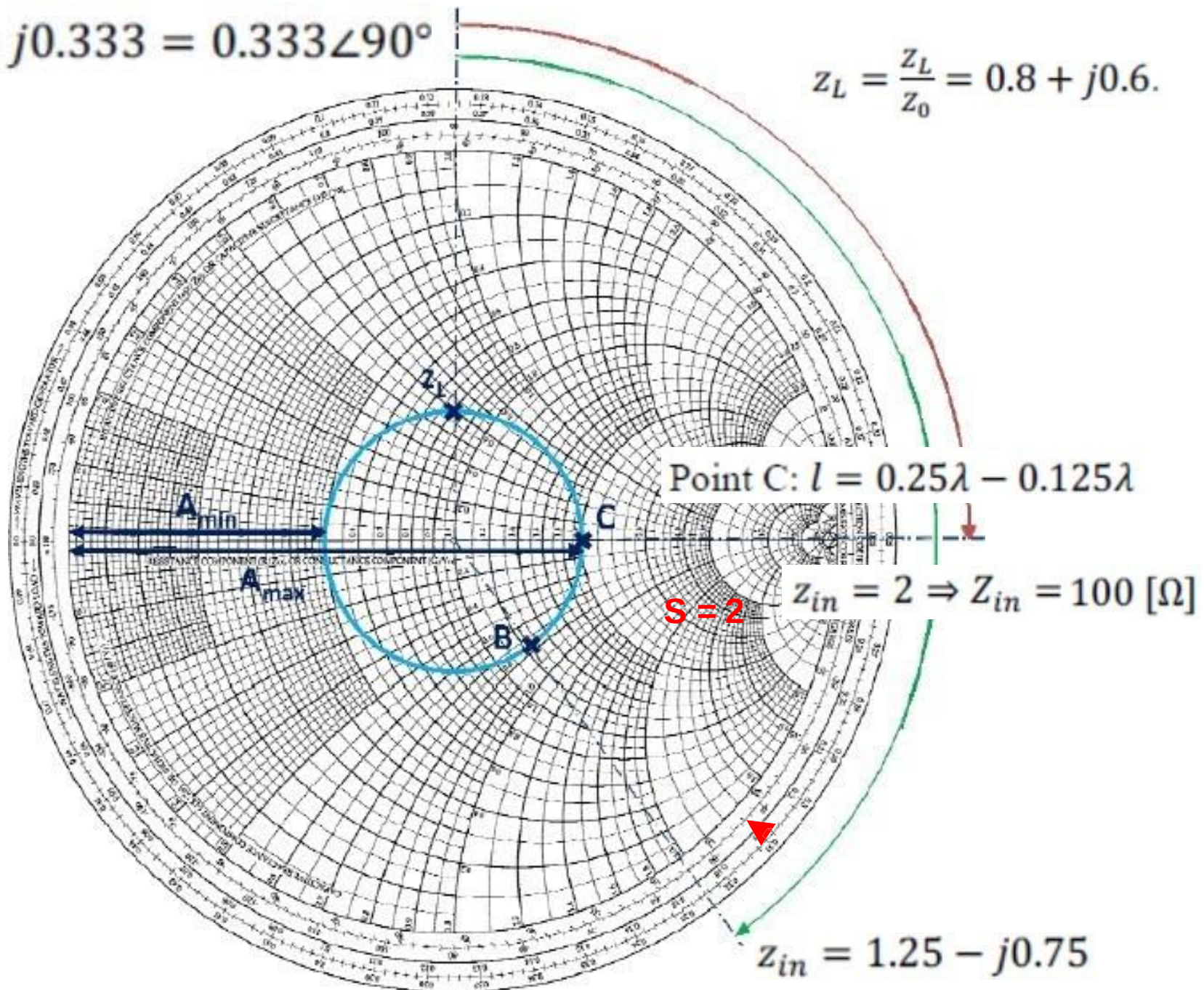


A lossless transmission line with a characteristic impedance of 50Ω is terminated in a load impedance $40 + j30 \Omega$. Using Smith Chart techniques, determine:

- a) The reflection coefficient Γ (magnitude and phase), and the VSWR
- b) The input impedance seen looking into the line if it is 0.2λ long.
- c) The length of line needed to make the input impedance look real (one solution is sufficient), and the associated resistance value.

a) $\Gamma = j0.333 = 0.333\angle 90^\circ$

$$z_L = \frac{Z_L}{Z_0} = 0.8 + j0.6.$$



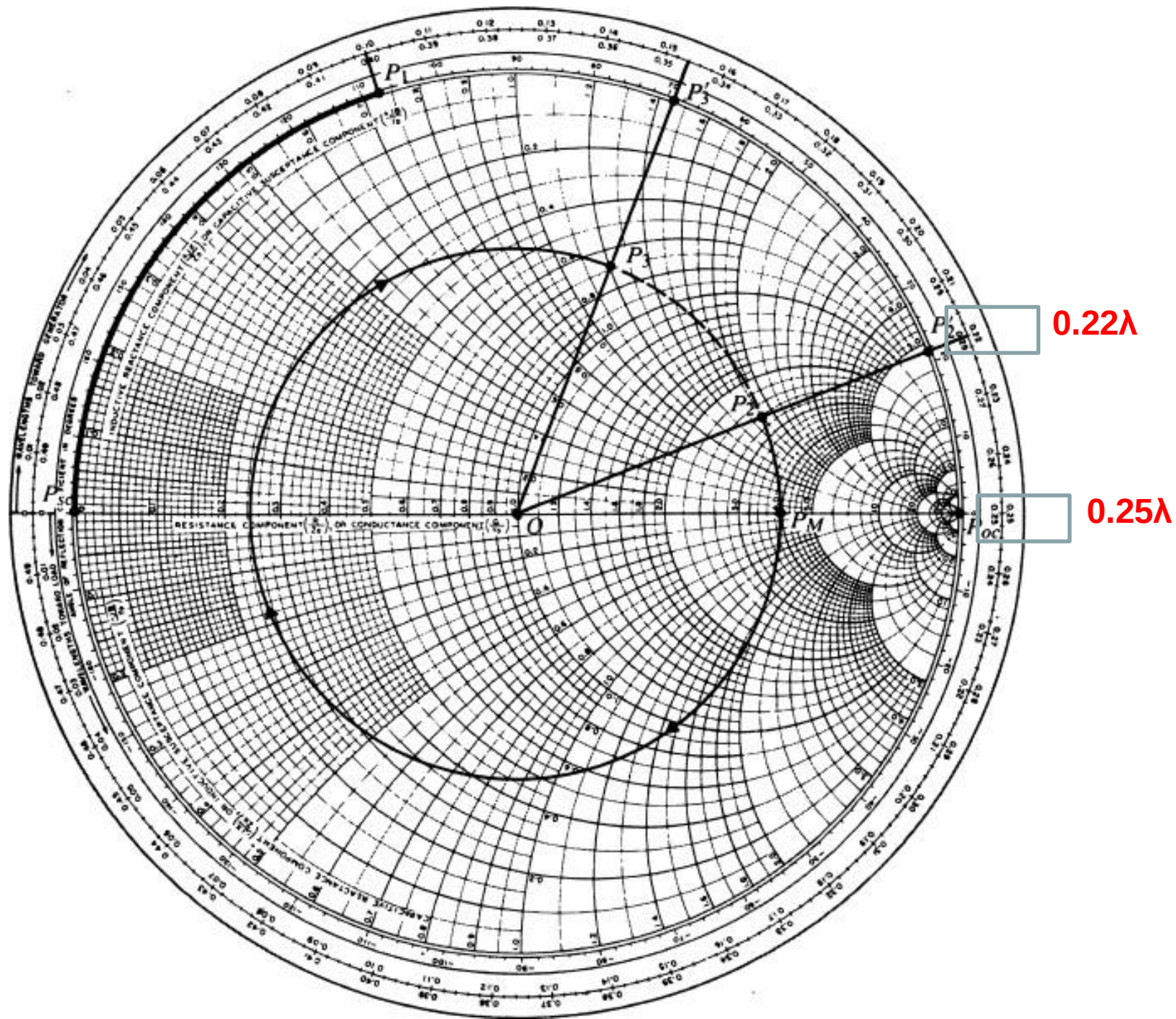
Example

EXAMPLE 9–14 A lossless transmission line of length 0.434λ and characteristic impedance $100\ \Omega$ is terminated in an impedance $260 + j180\ \Omega$. Find (a) the voltage reflection coefficient, (b) the standing-wave ratio, (c) the input impedance, and (d) the location of a voltage maximum on the line.

$$z' = 0.434\lambda,$$

$$R_0 = 100\ \Omega,$$

$$Z_L = 260 + j180$$



Why Impedance Matching



- **Malfunctioning of the source**
- **Breakdown (High VSWR)**
- **Frequency Pulling**

Sol: Impedance Matching